UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

03 07811

LH GENERAL PHYSICS

SUMMER EXAMINATIONS 2025

Time allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and two tables of physical constants and units that may be required will be found at the end of this question paper.

Answer **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

1. In this question, we will determine the maximum time that a perfectly sharp pencil can remain standing on its tip.

   (a) Approximate the pencil as a thin rod of uniform cross section, length l, and mass M. Show that the moment of inertia of the pencil about one of its ends is $I = l^2 M/3$. **[7]**

   (b) Suppose that you start the pencil perfectly upright, and it falls over under the influence of gravity. Let θ be the angle of the pencil, with $\theta = 0$ denoting an upright pencil. Show that for $\theta \ll 1$ the rate of change of θ is given by **[9]**

$$\dot{\theta} \approx \sqrt{\frac{3g}{2l}}\theta.$$

   (c) Suppose that the pencil starts with kinetic energy $\frac{1}{2}I\dot{\theta}^2 = \frac{1}{2}k_{\mathrm{B}}T$. From the expression given in section b, at what value of θ would the pencil attain this kinetic energy were it falling classically from a perfectly upright position? If we take the pencil to start at this position, as an approximation for its starting condition at temperature T, how long does it take to fall over? You may take the pencil to have fallen over when θ reaches 1. Set $T = 300$ K, $l = 0.15$ m, and $M = 0.02$ kg. **[7]**

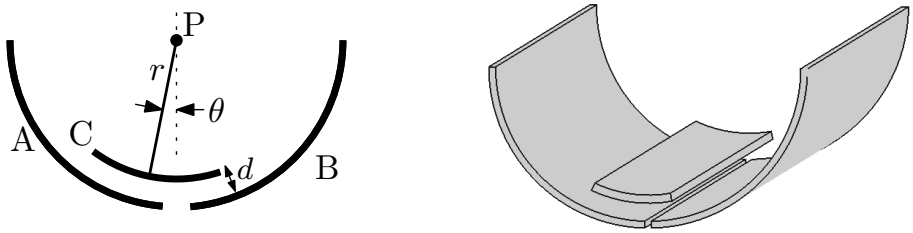
   (d) Now consider the $T \to 0$ limit, where thermal fluctuations are absent but there is still quantum uncertainty. At what value of uncertainty in θ, $\Delta\theta$, does the uncertainty in angular velocity of the pencil, $\Delta\dot{\theta}$, exceed the expected classical value of $\dot{\theta}$ from part b? If you start the pencil at $\theta = \Delta\theta$, about how long will it take to fall over? Again, set $l = 0.15$ m and $M = 0.02$ kg. **[7]**

2. (a) A parallel-plate capacitor consists of a pair of flat metal plates each having area A, separated by a distance d in vacuum. A charge Q has been transferred from one plate to the other by a battery of EMF V connected to the plates. Sketch the lines of electric field in the vicinity of the capacitor. **[3]**

(b) Explain why making d small compared to the dimensions of the plates simplifies calculations. Using this simplification, apply Gauss's law to show that the capacitance of the capacitor is given by $C = \epsilon_0 A/d$. **[5]**

(c) Show that the energy stored in the charged capacitor is given by the expression $\frac{1}{2}CV^2$. **[5]**

(d) The figure shows an electrometer intended to measure voltages. It consists of two fixed capacitor plates A and B which form parts of a cylinder, and a moving plate C, a narrow strip of a slightly smaller cylinder. C is supported by threads from the cylindrical axis P, and can rotate about this axis by a small angle θ, under the influence of electric fields and its weight mg. The system radius is r, and the spacing between the plates is d, where $d \ll r$.



Obtain expressions for the change in energy stored in the device when the potentials of A, B and C are V_A, V_B, and V_C and C moves from $\theta = 0$ to θ, and also for the energy that the batteries maintaining the potentials must supply as C moves to this position. **[8]**

(e) Hence or otherwise obtain an expression for the equilibrium angle θ as a function of V_A, V_B and V_C. **[4]**

(f) Show that if V_C is much larger than V_A and V_B, θ is a linear function of V_A and V_B, while if $V_B = V_C = 0$, θ depends on V_A^2. What might be an advantage of the latter case? **[5]**

3. (a) In a particular inertial reference frame, three events A, B and C occur at the following space-time coordinates:
A: $x = 6 \times 10^{12}\,\mathrm{m}$, $t = 0\,\mathrm{s}$,
B: $x = 5 \times 10^{11}\,\mathrm{m}$, $t = 3 \times 10^{3}\,\mathrm{s}$,
C: $x = 2 \times 10^{3}\,\mathrm{m}$, $t = 10^{4}\,\mathrm{s}$.
Calculate the space-time intervals beween these events and explain whether any of these events may have caused or influenced any of the other events. **[9]**

(b) Events A and B occured on two different planets. A spacecraft is travelling from planet A to planet B at a speed of $v = 0.9c$.

i. Calculate the distance between the planets, as measured from that spacecraft. State any assumptions that you make. **[3]**

ii. Explain whether the astronaut on this spacecraft concludes that there can be a causal relationship between the events A and B. **[2]**

(c) Event C is associated with the instantaneous and isotropic emission of $N_\gamma = 10^{27}$ photons, each of energy $E_\gamma = 10\,\mathrm{eV}$. Calculate the number of these photons that arrive per square metre of the planets on which events A and B occurred. State any assumptions that you make, and comment on your answer. **[6]**

(d) A radioactive isotope on the planet on which event B occurred has a half life $t_{1/2} = 60\,\mathrm{sec}$. This isotope emits a single gamma-ray each time one of its nuclei decays to a stable daughter nucleus. 100 moles of the isotope are distributed evenly over the surface of the planet.

i. Estimate how long, on average, it would take for the astronaut on the spaceship discussed in Part (b) to detect 25 gamma-rays from this isotope when it is at the mid-point of its journey between the two planets. You may assume that the astronaut has a device of surface area $1\,\mathrm{m}^2$ that is capable of detecting gamma-rays. State any other assumptions that you make. **[8]**

ii. Comment on the significance of detecting 25 gamma-rays. **[2]**

Formula Sheet

Vector notation is shown only in essential cases.

$$F = ma = \frac{\mathrm{d}p}{\mathrm{d}t}$$

$$v = u + at$$

$$v^2 = u^2 + 2as$$

$$\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p} = \boldsymbol{r} \times (m\boldsymbol{v})$$

$$\boldsymbol{\tau} = \boldsymbol{r} \times \boldsymbol{F}$$

$$\tau = I\frac{\mathrm{d}\omega}{\mathrm{d}t}$$

$$I = mr^2$$

$$I_P = I_{cm} + Mr^2$$

$$F_c = mv^2/r$$

$$a_r = \frac{v^2}{r} = \omega^2 r$$

$$\boldsymbol{F} = -\boldsymbol{\nabla}U \quad \text{or} \quad -\frac{\mathrm{d}U}{\mathrm{d}x}$$

$$K = \frac{1}{2}I\omega^2$$

$$\omega_0 = 2\pi f_0 = \sqrt{\frac{k_s}{m}}$$

$$F_G = \frac{Gm_1m_2}{r^2}$$

$$U = mgh$$

$$\boldsymbol{E} = \frac{Q}{4\pi\epsilon_0 r^2} = -\boldsymbol{\nabla}V$$

$$V = -\frac{Q}{4\pi\epsilon_0 r}$$

$$\mathrm{d}\boldsymbol{B} = \frac{\mu_0 I\,\mathrm{d}\boldsymbol{l} \times \hat{\boldsymbol{r}}}{4\pi r^2}$$

$$\boldsymbol{\nabla} \times \boldsymbol{B} = \mu_0\left(\boldsymbol{J} + \frac{\partial \boldsymbol{D}}{\partial t}\right)$$

$$\boldsymbol{\nabla} \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{E} = \frac{\rho}{\epsilon_0}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$$

$$\Phi = \int_S \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{A}$$

$$\epsilon = -\frac{\partial \Phi}{\partial t}$$

$$\oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{A} = \frac{Q}{\epsilon_0}$$

$$\oint \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{l} = \mu_0 \boldsymbol{I}$$

$$\boldsymbol{F} = q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B})$$

$$\boldsymbol{B} = \mu_0(\boldsymbol{H} + \boldsymbol{M})$$

$$\boldsymbol{D} = \epsilon_0\boldsymbol{E} + \boldsymbol{P}$$

$$u = \frac{1}{2}\epsilon_0 E^2$$

$$u = \frac{1}{2}\frac{B^2}{\mu_0}$$

$$R = \rho L/A$$

$$P = VI = I^2R = V^2/R$$

$$C = Q/V; \;\; C = \epsilon_0 a/d$$

$$\Phi = LI$$

$$\Phi_2 = M_{21}I_1$$

$$U = \frac{Q^2}{2C}$$

$$U = \frac{1}{2}LI^2$$

$$PV = nRT$$

$$C = \frac{\mathrm{d}Q}{\mathrm{d}T}$$

$$\gamma = \frac{C_P}{C_V}$$

$$PV^\gamma = K$$

$$v = \sqrt{\frac{\gamma RT}{m}}$$

$$\mathrm{d}U = đW + đQ$$

$$S = k_B \ln \Omega$$

$$\mathrm{d}S \geq \frac{đQ}{T}$$

$$P(E_n) = \frac{g_n}{Z} \exp\left(-\frac{E_n}{k_B T}\right)$$

$$\dot{Q} = kA\boldsymbol{\nabla}T \quad \text{or} \quad kA\frac{\mathrm{d}T}{\mathrm{d}x}$$

$$\dot{Q} = \epsilon\sigma A T^4$$

$$\lambda_{\max} T = K$$

$$v_p = f\lambda = \frac{\omega}{k}$$

$$v_g = \frac{\partial\omega}{\partial k}$$

$$k = 2\pi/\lambda$$

$$\omega = 2\pi f$$

$$Y = Y_0 \sin(kx - \omega t)$$

$$Y = Y_0 \exp i(kx - \omega t)$$

$$v = \sqrt{Y/\rho} \quad \text{or} \quad \sqrt{T/\mu}$$

$$\frac{f_{\text{rec}}}{f_{\text{src}}} = \frac{v \pm v_{\text{rec}}}{v \pm v_{\text{src}}}$$

$$2d \sin\theta = n\lambda$$

$$\gamma = \frac{1}{\sqrt{1-\beta^2}}$$

$$x' = \gamma(x - ut)$$

$$t' = \gamma(t - ux/c^2)$$

$$v' = \frac{v-u}{1 - uv/c^2}$$

$$p = \gamma m_0 v$$

$$E^2 = p^2c^2 + m_0^2c^4$$

$$\frac{f_{\text{rec}}}{f_{\text{src}}} = \sqrt{\frac{1-\beta}{1+\beta}}$$

$$E = hf$$

$$p = \frac{h}{\lambda}$$

$$N = N_0 \exp(-t/\tau) = N_0 \exp(-\lambda t)$$

$$\Delta p \cdot \Delta x \geq \hbar$$

$$\Delta E \cdot \Delta t \geq \hbar$$

$$\hat{\boldsymbol{p}} = -i\hbar\boldsymbol{\nabla}$$

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(r)\psi = E\psi$$

$$c = 1/\sqrt{\mu_0\epsilon_0}$$

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

$$n = c/v$$

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\sin\theta = 1.22\,\lambda/d$$

$$d\sin\theta = m\lambda$$

$$l = l_0(1 + \alpha\,\Delta T)$$

$$E = \frac{F/a}{\Delta l/l}$$

$$\gamma = F/d = W/a$$

$$p = \rho g h$$

$$F_{\text{up}} = (\rho_2 - \rho_1)gv$$

Astrophysical Constants and Units

Astronomical Unit	AU	1.50×10^{11} m
Parsec	pc	3.1×10^{16} m
Tropical year	y	365.242 mean solar days
Solar luminosity	$L_\odot$	3.84×10^{26} W
Absolute bolometric magnitude of the Sun	$M_{\rm bol}$	$+4^{\rm m}75$
Bolometric correction for the Sun	BC	$-0^{\rm m}08$
Apparent visual magnitude of the Sun	$m_v(\odot)$	$-26^{\rm m}74$
Solar constant	f	1.36×10^3 W m^{-2}
Solar mass	$M_\odot$	1.989×10^{30} kg
Solar radius	$R_\odot$	6.95×10^8 m
Solar effective temperature	$T_{\rm eff}$	5777 K
Wien constant	b	2.898×10^{-3} m K
Hubble constant	H_o	70 km s^{-1} Mpc^{-1}
Distance of the Sun from galactic centre		8.3 kpc
Earth mass	$M_\oplus$	5.972×10^{24} kg
Earth radius	$R_\oplus$	6371 km
Jupiter mass	$M_{\rm Jup}$	1.898×10^{27} kg
Jupiter radius	$R_{\rm Jup}$	71 492 km

Physical Constants and Units

Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	$273.15\,\mathrm{K}$
Avogadro constant	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\,\mathrm{m^{-1}}$
	$R_\infty hc$	$13.606\,\mathrm{eV}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
	$\hbar c$	$197.3\,\mathrm{MeV\,fm}$
Charge of proton	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Mass of electron	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Rest energy of electron		$0.511\,\mathrm{MeV}$
Mass of proton	m_p	$1.673 \times 10^{-27}\,\mathrm{kg}$
Rest energy of proton		$938.3\,\mathrm{MeV}$
One atomic mass unit	u	$1.66 \times 10^{-27}\,\mathrm{kg}$
Atomic mass unit energy equivalent		$931.5\,\mathrm{MeV}$
Electric constant	ϵ_0	$8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	7.297×10^{-3}= 1/137.0
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12}\,\mathrm{m}$
Bohr radius	a_0	$5.2918 \times 10^{-11}\,\mathrm{m}$
angstrom	Å	$10^{-10}\,\mathrm{m}$
barn	b	$10^{-28}\,\mathrm{m^2}$
torr (mm Hg at $0\,^\circ$C)	torr	$133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$)

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Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.